

Project Design Phase

Proposed Solution Template

Date: 01-07-2026

Team ID: SWUID2026200518

Project Name: Plugging into the Future: An Exploration of Electricity Consumption Patterns

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Electricity consumption data is often distributed across multiple reports and spreadsheets, making it difficult to analyze state-wise and region-wise trends. Decision-makers require an interactive platform to visualize electricity usage and support effective energy planning.
2	Idea / Solution Description	Develop an interactive Tableau dashboard integrated with a Flask web application that visualizes electricity consumption using maps, bar charts, pie charts, filters, and year-wise comparisons. The solution enables users to explore electricity consumption patterns efficiently.
3	Novelty / Uniqueness	The project combines Tableau's interactive visualization capabilities with a Flask-based web interface, providing an easy-to-use platform for analyzing electricity consumption. It presents geographical and analytical insights through dynamic dashboards rather than static reports.

4	Social Impact / Customer Satisfaction	The solution helps government agencies, researchers, utility companies, and students understand electricity usage patterns, supporting better energy planning, efficient resource allocation, and increased public awareness about electricity consumption.
5	Business Model (Revenue Model)	The platform can be offered as a subscription-based analytics service for utility companies and government organizations. Additional revenue can be generated through customized dashboards, consulting services, and enterprise deployments.
6	Scalability of the Solution	The architecture is scalable and can accommodate additional years of data, real-time electricity datasets, cloud deployment, predictive analytics, and integration with machine learning models without significant redesign.

Conclusion

The proposed solution provides an interactive and scalable platform for electricity consumption analysis. It simplifies complex datasets into meaningful visual insights, enabling faster decision-making and supporting future enhancements such as real-time analytics and predictive modeling.